Logarithm: 20 Hard-Level Questions

Chosen from the Solved Examples and the Exercise, with sub-questions, extra questions and step-by-step solutions

I worked through all 13 Solved Examples and all 92 Exercise questions in the chapter. The 20 questions below are the hardest ones. Each is the **lead question** of a group. Under it are **sub-questions** of the same family, so that the 20 groups together cover every pattern of the chapter. These patterns come up again and again in competitive exams in Bangladesh (BCS preliminary, bank recruitment, IBA MBA/BBA and university admission tests) and in India (CAT, MAT, XAT, IIFT, CDS, SSC, bank PO and railway exams).

Every question carries a tag that shows where it comes from:

- Exercise Q.22 a question from the Exercise of the PDF;
- Solved Ex. 10 a Solved Example of the PDF;
- Extra a question that is not in the PDF, added to fill a gap the PDF leaves.

A red tag such as M.B.A. 2007 shows the exam the PDF names for that question. The Exercise questions are multiple-choice in the PDF. Here they are asked directly, because the aim is to *work out* the answer, not to pick it. Every answer agrees with the PDF's answer key, except in the few places where the PDF has a misprint; those places are pointed out.

The solution comes straight after each question and each sub-question. A green **Concept** box explains the idea behind a step. A red **Trap** box shows the mistake that examiners hope you will make. When no base is written, $\log x$ means $\log_{10} x$, as in the PDF.

Where each pattern is covered

Q	Pattern	Lead question
1	Surd and decimal bases	Ex. Q.22
2	Rules for the base; unknown base	Ex. Q.21
3	Logs inside logs (nested)	Ex. Q.31
4	The identity $a^{\log_a x} = x$	Ex. Q.59
5	Long sums, telescoping, series	Ex. Q.36
6	Simplifying by prime factors	Ex. Q.39
7	Change of base: chains	Solved Ex. 10
8	Change of base: reciprocal sums	Solved Ex. 9
9	One log in terms of another	Ex. Q.53
10	Relations between unknown logs	Ex. Q.40
11	Equal powers: $a^x = b^y = \dots$	Ex. Q.69
12	Conditions and identities	Ex. Q.60
13	Equations with mixed bases	Ex. Q.70
14	Equations: domain, false roots	Ex. Q.55
15	Radicals, systems, 2^x inside a log	Ex. Q.71
16	Quadratics in disguise; AP of logs	Ex. Q.91
17	Values from $\log 2$, $\log 3$, $\log 7$	Ex. Q.80
18	Characteristic, mantissa, bar form	Ex. Q.84
19	Number of digits; zeros after the point	Ex. Q.90
20	Traps, comparisons, inequalities	Ex. Q.24

Part 0 · The Ideas Behind Every Solution

Read this once; every solution below refers back to it.

1. A logarithm is an exponent

The logarithm of x to the base a answers one question: *to what power must I raise a to get x?*

$$a^m = x \quad \Longleftrightarrow \quad m = \log_a x$$

For example, $\log_2 8 = 3$ because $2^3 = 8$, and $\log_{10} 0.01 = -2$ because $10^{-2} = \dfrac{1}{100}$. Whenever a log problem looks confusing, turn it back into this power form. Almost every solution in this document starts that way.

2. When does $\log_a x$ exist?

$\log_a x$ is a real number only when all three of these hold:

$$a > 0, \qquad a \neq 1, \qquad x > 0$$

- $x > 0$: a positive base raised to any power gives a positive number, so no power can give 0 or a negative number.
- $a \neq 1$: $1^m = 1$ for every m, so $\log_1 5$ has no answer and $\log_1 1$ has infinitely many.
- $a > 0$: a negative base gives non-real numbers for many powers; for example $(-4)^{\frac{1}{2}} = \sqrt{-4}$.

Figure 1 shows these rules as pictures.

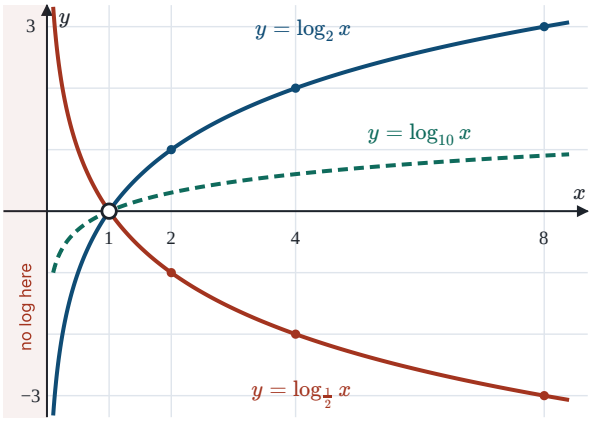


Figure 1. Graphs of $\log_2 x$, $\log_{10} x$ and $\log_{\frac{1}{2}} x$. All three meet at the ringed point $(1, 0)$.

Three facts can be read straight off the graph:

1. No curve enters the shaded strip, where $x \leq 0$. **The log of zero or of a negative number does not exist.**
2. Every curve passes through $(1, 0)$, because $a^0 = 1$. So $\log_a 1 = 0$ for every base.

3. If $a > 1$, the curve **rises**: a bigger x has a bigger log. If $0 < a < 1$, the curve **falls**: a bigger x has a smaller log. This decides which way an inequality points (Q.20).

3. Why the laws work

Let $\log_a x = m$ and $\log_a y = n$. Then $x = a^m$ and $y = a^n$, and each law follows from the laws of indices:

- **Product:** $xy = a^{m+n}$, so $\log_a(xy) = \log_a x + \log_a y$.
- **Quotient:** $\dfrac{x}{y} = a^{m-n}$, so $\log_a \dfrac{x}{y} = \log_a x - \log_a y$.
- **Power:** $x^p = a^{mp}$, so $\log_a x^p = p \log_a x$.
- **Change of base:** take $\log_b$ of $x = a^m$ to get $\log_b x = m \log_b a$, so
$$\log_a x = \frac{\log_b x}{\log_b a}.$$
- **Swap of base and number:** put $x = b$ in the change of base with base b, which gives $\log_a b = \dfrac{1}{\log_b a}$.
- **Undoing a log:** $a^{\log_a x} = a^m = x$.
- **Powers on both sides:** $\log_{a^q} x^p = \dfrac{p}{q} \log_a x$. In particular, $\log_{c^q} c^p = \dfrac{p}{q}$.

Part A · The Definition at Work

Questions 1–4: turning a log back into a power

Q.1 SURD AND DECIMAL BASES

Exercise Q.22

Find the value of $\log_{2\sqrt{3}}(1728)$.

SOLUTION

Step 1. Ask the defining question. $\log_{2\sqrt{3}} 1728$ asks: *to what power must $2\sqrt{3}$ be raised to give 1728?* So we try to write 1728 as a power of $2\sqrt{3}$.

Step 2. Get rid of the root. A square root disappears when it is squared:
$$\left(2\sqrt{3}\right)^2 = 2^2 \times \left(\sqrt{3}\right)^2 = 4 \times 3 = 12.$$

Step 3. Write 1728 as a power of 12. $12 \times 12 = 144$ and $144 \times 12 = 1728$, so $1728 = 12^3$. Now replace 12 by $\left(2\sqrt{3}\right)^2$:
$$1728 = 12^3 = \left[\left(2\sqrt{3}\right)^2\right]^3 = \left(2\sqrt{3}\right)^6.$$

Step 4. Read off the power.
$$\log_{2\sqrt{3}} 1728 = \log_{2\sqrt{3}} \left(2\sqrt{3}\right)^6 = 6.$$

Check by change of base. Since $2\sqrt{3} = 12^{\frac{1}{2}}$,
$$\log_{2\sqrt{3}} 1728 = \frac{\log 12^3}{\log 12^{\frac{1}{2}}} = \frac{3 \log 12}{\frac{1}{2} \log 12} = 6.$$

Answer: 6

CONCEPT: THE COMMON-BASE METHOD

Write the base and the number as powers of the **same** simple number c. Then
$$\log_{c^r} c^s = \frac{s}{r}, \quad \text{because} \quad (c^r)^{\frac{s}{r}} = c^s.$$
Here $c = 12$, the base is $12^{\frac{1}{2}}$ and the number is 12^3, so the answer is $3 \div \dfrac{1}{2} = 6$.

(a) Solved Ex. 3

If $\log_{\sqrt{8}} x = 3\dfrac{1}{3}$, find x.

First, $3\dfrac{1}{3}$ is a mixed number: $3\dfrac{1}{3} = 3 + \dfrac{1}{3} = \dfrac{10}{3}$. By the definition,
$$x = \left(\sqrt{8}\right)^{\frac{10}{3}}.$$
Write $\sqrt{8}$ as a power of 2: $\sqrt{8} = 8^{\frac{1}{2}} = \left(2^3\right)^{\frac{1}{2}} = 2^{\frac{3}{2}}$. Therefore
$$x = \left(2^{\frac{3}{2}}\right)^{\frac{10}{3}} = 2^{\frac{3}{2} \times \frac{10}{3}} = 2^5 = 32.$$

Answer: $x = 32$

TRAP

$3\dfrac{1}{3}$ means $3 + \dfrac{1}{3}$, not $3 \times \dfrac{1}{3}$. Reading it as 1 gives $x = \sqrt{8}$, which is wrong.

(b) Exercise Q.5 · M.A.T. 2005

Find $\log_{3\sqrt{2}} \left(\dfrac{1}{18}\right)$.

Square the base: $\left(3\sqrt{2}\right)^2 = 9 \times 2 = 18$. A fraction $\dfrac{1}{N}$ is N^{-1}, so
$$\frac{1}{18} = 18^{-1} = \left[\left(3\sqrt{2}\right)^2\right]^{-1} = \left(3\sqrt{2}\right)^{-2}.$$
$$\therefore \log_{3\sqrt{2}} \frac{1}{18} = -2.$$

Answer: -2

(c) Exercise Q.10

Find the logarithm of 0.00001 to the base 0.01.

Write both numbers as powers of 10. Count the decimal places: 0.01 has two, so $0.01 = 10^{-2}$; 0.00001 has five, so $0.00001 = 10^{-5}$. With the common-base method,
$$\log_{0.01} 0.00001 = \log_{10^{-2}} 10^{-5} = \frac{-5}{-2} = \frac{5}{2}.$$
Check: $(0.01)^{\frac{5}{2}} = \left(10^{-2}\right)^{\frac{5}{2}} = 10^{-5} = 0.00001$. ✓

Answer: $\dfrac{5}{2}$

TRAP: THE SIGN OF A LOG

Both numbers are less than 1, yet the log is **positive**. The sign depends on whether the number and the base lie on the **same side of 1**. Same side gives a positive log; opposite sides give a negative log. Part (d) shows the opposite case.

(d) Exercise Q.7

Find $\log_{0.01}(1000)$.

$0.01 = 10^{-2}$ is below 1 and $1000 = 10^3$ is above 1, so expect a negative answer:

$$\log_{10^{-2}} 10^3 = \frac{3}{-2} = -\frac{3}{2}.$$

Answer: $-\frac{3}{2}$

(e) Extra

Find $\log_{\sqrt[3]{4}} \left(\frac{1}{16\sqrt{2}}\right)$.

Everything is a power of 2. The base is $\sqrt[3]{4} = (2^2)^{\frac{1}{3}} = 2^{\frac{2}{3}}$. For the number, $16\sqrt{2} = 2^4 \times 2^{\frac{1}{2}} = 2^{\frac{9}{2}}$, so $\frac{1}{16\sqrt{2}} = 2^{-\frac{9}{2}}$. Hence

$$\log_{2^{\frac{2}{3}}} 2^{-\frac{9}{2}} = \frac{-\frac{9}{2}}{\frac{2}{3}} = -\frac{9}{2} \times \frac{3}{2} = -\frac{27}{4}.$$

Answer: $-\frac{27}{4}$

Q.2 RULES FOR THE BASE; UNKNOWN BASE

Exercise Q.21

Find the value of $\log_{\left(-\frac{1}{3}\right)} 81$. (The PDF offers -27, -4, 4 and 27.)

SOLUTION 1: THE METHOD THE EXAMINER EXPECTS

Let the value be x. By the definition,

$$\left(-\frac{1}{3}\right)^x = 81.$$

Now $81 = 3^4$. An even power hides the sign, so $3^4 = (-3)^4$. Also -3 is the reciprocal of $-\frac{1}{3}$, that is, $-3 = \left(-\frac{1}{3}\right)^{-1}$. Therefore

$$81 = (-3)^4 = \left[\left(-\frac{1}{3}\right)^{-1}\right]^4 = \left(-\frac{1}{3}\right)^{-4}.$$

Comparing powers gives $x = -4$. This is the PDF's answer.

SOLUTION 2: THE STRICT MATHEMATICAL VIEW

Part 0 says the base of a logarithm must be **positive** (and not 1). The base here is negative. For many powers a negative base does not even give a real number; for example, $\left(-\frac{1}{3}\right)^{\frac{1}{2}} = \sqrt{-\frac{1}{3}}$ is not real. So in real numbers, $\log_{\left(-\frac{1}{3}\right)} 81$ is **not defined**.

Answer: -4 (as the PDF expects)

HOW TO ANSWER IN THE EXAM

If the options contain -4 and no "not defined", choose -4. If "not defined" is offered, or in a written exam, say that the expression is not defined because the base is negative.

TRAP

$+4$ is the tempting wrong option: $\left(-\frac{1}{3}\right)^4 = \frac{1}{81}$, not 81. A negative power is needed to turn the fraction upside down.

(a) Exercise Q.14

If $\log_x \left(\frac{9}{16}\right) = -\frac{1}{2}$, find x.

Power form: $x^{-\frac{1}{2}} = \frac{9}{16}$. A negative power means "one over", and a power $\frac{1}{2}$ means a square root, so $x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$. Hence

$$\frac{1}{\sqrt{x}} = \frac{9}{16} \Rightarrow \sqrt{x} = \frac{16}{9} \Rightarrow x = \left(\frac{16}{9}\right)^2 = \frac{256}{81}.$$

This base is positive and not 1, so it is allowed.

Answer: $x = \frac{256}{81}$

CONCEPT: FREEING x FROM A POWER

If $x^{\frac{p}{q}} = N$, raise both sides to the reciprocal power $\frac{q}{p}$:

$$x = N^{\frac{q}{p}}, \quad \text{because} \quad \left(x^{\frac{p}{q}}\right)^{\frac{q}{p}} = x^1 = x.$$

In (a), the power is $-\frac{1}{2}$, so $x = \left(\frac{9}{16}\right)^{-2} = \frac{256}{81}$ in one step.

(b) Exercise Q.15

If $\log_x 4 = 0.4$, find x.

Write the decimal as a fraction: $0.4 = \frac{4}{10} = \frac{2}{5}$. Power form: $x^{\frac{2}{5}} = 4$. Raise both sides to the power $\frac{5}{2}$:

$$x = 4^{\frac{5}{2}} = \left(4^{\frac{1}{2}}\right)^5 = 2^5 = 32.$$

Answer: $x = 32$

(c) Exercise Q.18

If $\log_x(0.1) = -\frac{1}{3}$, find x.

Power form: $x^{-\frac{1}{3}} = \frac{1}{10}$. Raise both sides to the power -3:

$$x = \left(\frac{1}{10}\right)^{-3} = 10^3 = 1000.$$

Answer: $x = 1000$

(d) Exercise Q.16 & Q.19 N.M.A.T. 2006

Find x in each case.

(i) $\log_{10000} x = -\frac{1}{4}$ (ii) $\log_{32} x = 0.8$

Here the base is known and the number is unknown, so the power form gives x at once.

(i) $x = 10000^{-\frac{1}{4}} = \left(10^4\right)^{-\frac{1}{4}} = 10^{-1} = \frac{1}{10}$.

(ii) $0.8 = \frac{4}{5}$, so $x = 32^{\frac{4}{5}} = \left(2^5\right)^{\frac{4}{5}} = 2^4 = 16$.

Answer: (i) $\frac{1}{10}$ (ii) 16

(e) Extra

Solve $\log_x(6 - x) = 2$.

Power form: $x^2 = 6 - x$, so $x^2 + x - 6 = 0$. Factorise: we need two numbers whose product is -6 and sum is 1; they are 3 and -2.

$$(x + 3)(x - 2) = 0 \Rightarrow x = -3 \text{ or } x = 2.$$

Now apply the rules for the base. $x = -3$ would be a negative base, so reject it. For $x = 2$: the base 2 is positive and not 1, and the number $6 - 2 = 4$ is positive. Indeed $\log_2 4 = 2$. ✓

Answer: $x = 2$ only

(f) Extra

For which real values of x is $\log_{(x-2)}(10 - x)$ defined?

All three conditions of Part 0 must hold together:

- base positive: $x - 2 > 0$, so $x > 2$;
- base not 1: $x - 2 \neq 1$, so $x \neq 3$;
- number positive: $10 - x > 0$, so $x < 10$.

Answer: $2 < x < 10$ and $x \neq 3$

Q.3 LOGS INSIDE LOGS

Exercise Q.31

Find the value of

$$\log_2 \left[\log_2 \left\{\log_4 \left(\log_4 256^4\right)\right\}\right].$$

SOLUTION

Work from the **inside out**, like peeling an onion. Never touch an outer log until everything inside it has become a plain number.

Step 1. Simplify the innermost number. $4 \times 4 = 16$ and $16 \times 16 = 256$, so $256 = 4^4$. Then

$$256^4 = \left(4^4\right)^4 = 4^{16}.$$

Step 2. Innermost log. $\log_4 4^{16} = 16$.

Step 3. Next log. $\log_4 16 = \log_4 4^2 = 2$.

Step 4. Next log. $\log_2 2 = 1$.

Step 5. Outer log. $\log_2 1 = 0$, because $2^0 = 1$.

$\log_4 256^4 = \log_4 4^{16} = 16$

↓ apply $\log_4$

$\log_4 16 = 2$

↓ apply $\log_2$

$\log_2 2 = 1$

↓ apply $\log_2$

$\log_2 1 = 0$

Figure 2. Peeling a nested log from the inside out. Each box gives a plain number before the next log acts on it.

Answer: 0

TRAP

Do not work out $256^4 = 4\,294\,967\,296$. Turn it into a power of the base (here 4^{16}) and the logs fall away one by one.

(a) Solved Ex. 6 · Exercise Q.28 · M.B.A. 2005, 2007

Find x if

$$\log_2 \left[\log_3 \left(\log_2 x\right)\right] = 1.$$

Now the unknown is deep inside, so peel from the **outside in**. Each step turns one log into a power with the rule $\log_a A = k \Rightarrow A = a^k$.

$$\begin{aligned} \log_2 \left[\log_3(\log_2 x)\right] = 1 &\Rightarrow \log_3(\log_2 x) = 2^1 = 2 \\ &\Rightarrow \log_2 x = 3^2 = 9 \\ &\Rightarrow x = 2^9 = 512. \end{aligned}$$

Answer: $x = 512$

NOTE ON THE PDF

The printed solution of Solved Ex. 6 stops at $\log_2 x = 9$. The question asks for x, so one more step is needed: $x = 2^9 = 512$. This agrees with Exercise Q.28, option (d).

(b) Exercise Q.30

Find $\log_2 \log_2 \log_3 \log_3 27^3$.

Inside out: $27^3 = \left(3^3\right)^3 = 3^9$, then

$$\log_3 3^9 = 9, \quad \log_3 9 = 2, \quad \log_2 2 = 1, \quad \log_2 1 = 0.$$

Answer: 0

(c) Exercise Q.29

Find $\log_{10} \log_{10} \log_{10} \left(10^{10^{10}}\right)$.

Here $10^{10^{10}}$ means 10 raised to the power 10^{10}. Inside out:

$$\log_{10} 10^{10^{10}} = 10^{10}, \quad \log_{10} 10^{10} = 10, \quad \log_{10} 10 = 1.$$

Answer: 1

NOTE ON THE PDF

The question is printed as 10^{1010}, which is a misprint for $10^{10^{10}}$, as the PDF's own solution shows. Read literally, $\log\log 1010 \approx \log 3.004 \approx 0.478$, which is not among the options.

(d) Exercise Q.8 · M.B.A. 2010

Find $\left[\log_{10}\left(5 \log_{10} 100\right)\right]^2$.

Inside first: $\log_{10} 100 = 2$, so $5 \log_{10} 100 = 10$. Then $\log_{10} 10 = 1$, and $1^2 = 1$.

Answer: 1

TRAP

$[\log A]^2$ is the **square of a log**. It is not $\log A^2 = 2 \log A$.

(e) Extra

(i) For which x is $\log_2 (\log_3 x)$ defined? (ii) Solve $\log_2 \log_3 \log_4 x = 0$.

(i) There are two layers of conditions. The inner log needs $x > 0$. The outer log needs its input to be positive: $\log_3 x > 0$. From Figure 1, a log with base above 1 is positive exactly when the number is above 1. So $x > 1$, which also covers $x > 0$.

(ii) Outside in, using $\log_a A = k \Rightarrow A = a^k$:

$$\begin{aligned} \log_2 (\log_3 \log_4 x) = 0 &\Rightarrow \log_3 \log_4 x = 2^0 = 1 \\ &\Rightarrow \log_4 x = 3^1 = 3 \\ &\Rightarrow x = 4^3 = 64. \end{aligned}$$

Answer: (i) $x > 1$ (ii) $x = 64$

Q.4 THE IDENTITY $a^{\log_a x} = x$

Exercise Q.59

Find the value of $16^{\log_4 5}$.

SOLUTION

The idea. $\log_a x$ is *the power of a that gives x*. So raising a to that power gives x back: $a^{\log_a x} = x$. For example, $4^{\log_4 5} = 5$.

Step 1. Make the base match the log's base. The base here is 16 but the log has base 4, so the identity cannot be used yet. Write $16 = 4^2$:

$$16^{\log_4 5} = \left(4^2\right)^{\log_4 5} = 4^{2\log_4 5}.$$

Step 2. Move the 2 inside the log with the power law, $2\log_4 5 = \log_4 5^2 = \log_4 25$:

$$4^{2\log_4 5} = 4^{\log_4 25} = 25.$$

Another way. Let $y = 16^{\log_4 5}$ and take $\log_4$ of both sides:

$$\log_4 y = \log_4 5 \times \log_4 16 = 2\log_4 5 = \log_4 25,$$

so $y = 25$.

Answer: 25

TRAP

The answer is not 5. That would be the value of $4^{\log_4 5}$; here the base is $16 = 4^2$, which squares the result.

(a) Solved Ex. 2(iii)

Find $36^{\log_6 4}$.

$$36^{\log_6 4} = \left(6^2\right)^{\log_6 4} = 6^{2\log_6 4} = 6^{\log_6 16} = 16.$$

Answer: 16

(b) Extra

Find $49^{1-\log_7 2}$.

Split the power with $a^{m-n} = \dfrac{a^m}{a^n}$:

$$49^{1-\log_7 2} = \frac{49}{49^{\log_7 2}} = \frac{49}{7^{2\log_7 2}} = \frac{49}{7^{\log_7 4}} = \frac{49}{4}.$$

Answer: $\dfrac{49}{4}$

(c) Extra

Find the value of

$$81^{\frac{1}{\log_5 3}}.$$

Swap base and number: $\dfrac{1}{\log_5 3} = \log_3 5$. Then $81 = 3^4$ gives

$$81^{\log_3 5} = \left(3^4\right)^{\log_3 5} = 3^{\log_3 5^4} = 5^4 = 625.$$

Answer: 625

(d) Extra

Show that $3^{\log 5} = 5^{\log 3}$.

Hence find $3^{\log 5} - 5^{\log 3}$.

Take the log of each number:

$$\log\left(3^{\log 5}\right) = \log 5 \times \log 3, \qquad \log\left(5^{\log 3}\right) = \log 3 \times \log 5.$$

The two logs are equal. Figure 1 shows that each log value comes from exactly one positive number, so the numbers themselves are equal. Hence $3^{\log 5} - 5^{\log 3} = 0$.

Answer: 0

CONCEPT

This is property 9 of the PDF: $x^{\log_a y} = y^{\log_a x}$. The base and the number inside the log can swap places without changing the value.

Part B · Laws, Simplification and Change of Base

Questions 5–8: long expressions that collapse to a single number

Q.5 LONG SUMS, TELESCOPING, SERIES

Exercise Q.36

Find the value of

$$\log_{10} 1\frac{1}{2} + \log_{10} 1\frac{1}{3} + \log_{10} 1\frac{1}{4} + \cdots$$

up to 198 terms.

SOLUTION

Step 1. Turn the mixed numbers into fractions. $1\dfrac{1}{2} = \dfrac{3}{2}$, $1\dfrac{1}{3} = \dfrac{4}{3}$, $1\dfrac{1}{4} = \dfrac{5}{4}$, and so on. The 1st term has denominator 2, the 2nd has 3, so the 198th has denominator 199. The last term is therefore $1\dfrac{1}{199} = \dfrac{200}{199}$.

Step 2. A sum of logs is the log of a product.

$$\text{Sum} = \log_{10}\left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \cdots \times \frac{200}{199}\right).$$

Step 3. Cancel. Each numerator cancels the next denominator. Only the first denominator (2) and the last numerator (200) survive:

$$\frac{\cancel{3}}{2} \times \frac{\cancel{4}}{\cancel{3}} \times \frac{\cancel{5}}{\cancel{4}} \times \cdots \times \frac{200}{\cancel{199}} = \frac{200}{2} = 100.$$

Step 4. $\log_{10} 100 = \log_{10} 10^2 = 2$.

Answer: 2

CONCEPT: TELESCOPING

A long sum of logs of fractions, where each top matches the next bottom, folds up like a telescope into one fraction: first bottom under last top.

TRAP

The last term is $\dfrac{200}{199}$, not $\dfrac{199}{198}$. Always find the last term from the pattern "term k has denominator $k+1$".

(a) Exercise Q.26

Find $\log_{10}\left(10 \times 10^2 \times 10^3 \times \cdots \times 10^9\right)$.

When powers of the same base are multiplied, their exponents are added:

$$10^1 \times 10^2 \times \cdots \times 10^9 = 10^{1+2+\cdots+9}.$$

The sum of the first n counting numbers is $\dfrac{n(n+1)}{2}$, so $1+2+\cdots+9=\dfrac{9\times 10}{2}=45$. Hence the value is $\log_{10}10^{45}=45$.

Answer: 45

(b) Exercise Q.92 CDS 2012

If $\log_{10}a=p$ and $\log_{10}b=q$, find $\log_{10}(a^pb^q)$.

$$\begin{aligned}\log(a^pb^q)&=\log a^p+\log b^q\\&=p\log a+q\log b\\&=p\times p+q\times q=p^2+q^2.\end{aligned}$$

Answer: p^2+q^2

(c) Extra

If $\log_2(2\times 4\times 8\times\cdots\times 2^n)=55$, find n.

The product is

$$2^1\times 2^2\times\cdots\times 2^n=2^{1+2+\cdots+n}=2^{\frac{n(n+1)}{2}}.$$

Taking $\log_2$ of both sides:

$$\begin{aligned}\frac{n(n+1)}{2}=55 &\Rightarrow n^2+n-110=0\\&\Rightarrow (n-10)(n+11)=0.\end{aligned}$$

The number of factors cannot be negative, so $n=10$.

Answer: $n=10$

(d) Extra

Simplify

$$\log\left(1-\frac{1}{2^2}\right)+\log\left(1-\frac{1}{3^2}\right)+\cdots+\log\left(1-\frac{1}{100^2}\right).$$

Step 1. Factorise one bracket. Using $k^2-1=(k-1)(k+1)$,

$$1-\frac{1}{k^2}=\frac{k^2-1}{k^2}=\frac{k-1}{k}\times\frac{k+1}{k}.$$

Step 2. Multiply all brackets ($k=2$ to 100). Each half telescopes on its own:

$$\begin{aligned}\frac{1}{2}\times\frac{2}{3}\times\cdots\times\frac{99}{100}&=\frac{1}{100},\\\frac{3}{2}\times\frac{4}{3}\times\cdots\times\frac{101}{100}&=\frac{101}{2}.\end{aligned}$$

Step 3. The product is $\dfrac{1}{100}\times\dfrac{101}{2}=\dfrac{101}{200}$, so the sum is $\log\dfrac{101}{200}$ (about -0.2967).

Answer: $\log\dfrac{101}{200}$

(e) Extra

Find $\log_{\sqrt{2}}\left(2\times\sqrt{2}\times\sqrt[4]{2}\times\sqrt[8]{2}\times\cdots\right)$, the product going on for ever.

Write every factor as a power of 2: $2^1\times 2^{\frac12}\times 2^{\frac14}\times 2^{\frac18}\times\cdots$. The exponents form an infinite geometric series with first term 1 and ratio $\dfrac12$. Such a series adds up to $\dfrac{\text{first term}}{1-\text{ratio}}$:

$$1+\frac12+\frac14+\cdots=\frac{1}{1-\frac12}=2.$$

So the product is $2^2=4$, and $\log_{\sqrt{2}}4=\log_{2^{\frac12}}2^2=\dfrac{2}{\frac12}=4$.

Answer: 4

Q.6 SIMPLIFYING BY PRIME FACTORS

Exercise Q.39

Find the value of

$$\log 2+16\log\frac{16}{15}+12\log\frac{25}{24}+7\log\frac{81}{80}.$$

SOLUTION

Step 1. Break every number into primes. All the numbers here are built from 2, 3 and 5 only:

$$\frac{16}{15}=\frac{2^4}{3\times 5},\quad \frac{25}{24}=\frac{5^2}{2^3\times 3},\quad \frac{81}{80}=\frac{3^4}{2^4\times 5}.$$

Step 2. Move each coefficient up as a power ($k\log x=\log x^k$), and combine into one log, $\log N$, where

$$N=2\times\left(\frac{2^4}{3\cdot 5}\right)^{16}\left(\frac{5^2}{2^3\cdot 3}\right)^{12}\left(\frac{3^4}{2^4\cdot 5}\right)^{7}.$$

Step 3. Keep a tally of each prime. A prime on top counts plus, a prime underneath counts minus, and each count is multiplied by the outside power:

Factor	power of 2	power of 3	power of 5
2	$+1$	0	0
$\left(\frac{16}{15}\right)^{16}$	$+64$	-16	-16
$\left(\frac{25}{24}\right)^{12}$	-36	-12	$+24$
$\left(\frac{81}{80}\right)^{7}$	-28	$+28$	-7
Total	1	0	1

Step 4. $N=2^1\times 3^0\times 5^1=10$, so the expression is $\log 10=1$.

Second method. Put $\log 2=a$ and $\log 3=b$. Then $\log 5=\log\dfrac{10}{2}=1-a$, and the expression becomes

$$\begin{aligned}&a+16(5a-b-1)\\&+12(2-5a-b)\\&+7(4b-3a-1).\end{aligned}$$

The a terms give $1+80-60-21=0$, the b terms give $-16-12+28=0$, and the constants give $-16+24-7=1$. The value is 1 again.

Answer: 1

CONCEPT

When a log expression is full of "ugly" fractions, factorise every number into primes and count the powers. The coefficients are chosen so that almost everything cancels.

(a) Solved Ex. 5

Simplify $\log\dfrac{75}{16}-2\log\dfrac{5}{9}+\log\dfrac{32}{243}$.

First $2\log\dfrac{5}{9}=\log\dfrac{25}{81}$. Subtracting a log means dividing, which is multiplying by the upside-down fraction:

$$\log\left(\frac{75}{16}\times\frac{81}{25}\times\frac{32}{243}\right).$$

Pair the numbers: $\frac{75}{25}=3$, $\frac{81}{243}=\frac{1}{3}$ and $\frac{32}{16}=2$. The product is $3\times\frac{1}{3}\times 2=2$.

Answer: $\log 2$

(b) Exercise Q.34

Find

$$\log\frac{26}{51}+\log\frac{119}{91}-\log\frac{13}{32}-\log\frac{64}{39}.$$

Group the plus terms and the minus terms, and factorise. Here $26=2\cdot 13$, $51=3\cdot 17$, $119=7\cdot 17$, $91=7\cdot 13$ and $39=3\cdot 13$, so

$$\frac{26}{51}\times\frac{119}{91}=\frac{2\cdot 13\cdot 7\cdot 17}{3\cdot 17\cdot 7\cdot 13}=\frac{2}{3},$$
$$\frac{13}{32}\times\frac{64}{39}=\frac{13\cdot 2\cdot 32}{32\cdot 3\cdot 13}=\frac{2}{3}.$$

So the value is $\log\frac{2}{3}-\log\frac{2}{3}=0$.

Answer: 0

(c) Exercise Q.37 I.I.F.T. 2005

Find $\log\frac{9}{14}-\log\frac{15}{16}+\log\frac{35}{24}$.

$$\log\left(\frac{9}{14}\times\frac{16}{15}\times\frac{35}{24}\right)=\log\frac{3^2\cdot 2^4\cdot 5\cdot 7}{2\cdot 7\cdot 3\cdot 5\cdot 2^3\cdot 3}.$$

Top and bottom are both $2^4\cdot 3^2\cdot 5\cdot 7=5040$, so the value is $\log 1=0$.

Answer: 0

(d) Exercise Q.62 M.B.A. 2006

Find $\log\frac{a^2}{bc}+\log\frac{b^2}{ac}+\log\frac{c^2}{ab}$.

$$\log\left(\frac{a^2}{bc}\times\frac{b^2}{ac}\times\frac{c^2}{ab}\right)=\log\frac{a^2b^2c^2}{a^2b^2c^2}=\log 1=0.$$

Answer: 0

(e) Exercise Q.35 & Q.38 M.B.A. 2002

Find

(i) $\frac{1}{3}\log_{10}125-2\log_{10}4+\log_{10}32$;

(ii) $2\log_{10}5+\log_{10}8-\frac{1}{2}\log_{10}4$.

(i) $\frac{1}{3}\log 125=\log 125^{\frac{1}{3}}=\log 5$ and $2\log 4=\log 16$, so the value is

$$\log\frac{5\times 32}{16}=\log 10=1.$$

(ii) $2\log 5=\log 25$ and $\frac{1}{2}\log 4=\log 2$, so the value is

$$\log\frac{25\times 8}{2}=\log 100=2.$$

Answer: (i) 1 (ii) 2

TRAP

Move every coefficient up as a power **before** combining. Writing $2\log 5+\log 8=\log(2\times 5\times 8)$ is a common slip.

Q.7 CHANGE OF BASE: CHAINS

Solved Ex. 10 M.A.T. 2006

If

$$\log_a b=\frac{1}{2},\quad \log_b c=\frac{1}{3}\quad\text{and}\quad \log_c a=\frac{k}{5},$$

find the value of k.

SOLUTION

Step 1. Write each log with a common base (base 10), using the change-of-base rule:

$$\log_a b=\frac{\log b}{\log a},\quad \log_b c=\frac{\log c}{\log b},\quad \log_c a=\frac{\log a}{\log c}.$$

Step 2. Multiply the three. Every log on top also appears underneath, so everything cancels:

$$\frac{\log b}{\log a}\times\frac{\log c}{\log b}\times\frac{\log a}{\log c}=1.$$

Step 3. Use the given values.

$$\frac{1}{2}\times\frac{1}{3}\times\frac{k}{5}=1\ \Rightarrow\ \frac{k}{30}=1\ \Rightarrow\ k=30.$$

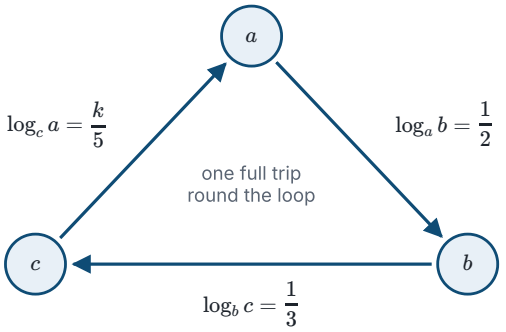


Figure 3. Going $a\to b\to c\to a$ ends where it began, so the product of the three logs is $\log_a a=1$.

Answer: $k=30$

CONCEPT: THE CHAIN RULE

When the number of one log is the base of the next, the middle letter cancels, just like fractions:

$$\log_a b\times\log_b c=\log_a c\quad\left(\text{like }\frac{b}{a}\times\frac{c}{b}=\frac{c}{a}\right).$$

A chain that returns to its starting letter gives $\log_a a=1$.

(a) Exercise Q.58

Find

$$(\log_3 4)(\log_4 5)(\log_5 6)(\log_6 7)(\log_7 8)(\log_8 9).$$

A perfect chain: 4, 5, 6, 7 and 8 all cancel, leaving the first base and the last number:

$$\log_3 9=\log_3 3^2=2.$$

Answer: 2

(b) Solved Ex. 4(i)

Find $\log_5 3\times\log_{27}25$.

This is not a ready-made chain (3 is not 27, and 25 is not 5), so change to base 10 and use the power law:

$$\frac{\log 3}{\log 5} \times \frac{\log 5^2}{\log 3^3} = \frac{\log 3}{\log 5} \times \frac{2\log 5}{3\log 3} = \frac{2}{3}.$$

Answer: $\frac{2}{3}$

(c) Solved Ex. 4(ii)

Find $\log_9 27 - \log_{27} 9$.

Both are powers of 3, so use $\log_{c^r} c^s = \frac{s}{r}$:

$$\log_{3^2} 3^3 - \log_{3^3} 3^2 = \frac{3}{2} - \frac{2}{3} = \frac{9-4}{6} = \frac{5}{6}.$$

Answer: $\frac{5}{6}$

(d) Exercise Q.51 & Q.52

Find each product.

(i) $(\log_5 3)(\log_3 625)$ (ii) $(\log_5 5)(\log_4 9)(\log_3 2)$

(i) A chain: $\log_5 625 = \log_5 5^4 = 4$.

(ii) $\log_5 5 = 1$. Next, $\log_4 9 = \log_{2^2} 3^2 = \frac{2}{2}\log_2 3 = \log_2 3$. Finally $\log_2 3 \times \log_3 2 = 1$, because the two are reciprocals. So the product is 1.

Answer: (i) 4 (ii) 1

(e) Extra

If $\log_2 3 \cdot \log_3 4 \cdot \log_4 5 \cdots \log_n (n+1) = 10$, find n.

The chain collapses to the first base and the last number: $\log_2(n+1) = 10$. So $n+1 = 2^{10} = 1024$.

Answer: $n = 1023$

Q.8 CHANGE OF BASE: RECIPROCAL SUMS

Solved Ex. 9 M.A.T. 2005

Simplify

$$\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)}.$$

SOLUTION

Step 1. Flip each term. Since $\frac{1}{\log_a b} = \log_b a$,

$$\frac{1}{\log_{xy}(xyz)} = \log_{xyz}(xy),$$

and in the same way for the other two terms. Now all three logs have the **same base** xyz, so they can be added.

Step 2. Add.

$$\begin{aligned}\text{Sum} &= \log_{xyz}(xy \cdot yz \cdot zx) \\ &= \log_{xyz}(x^2y^2z^2) \\ &= \log_{xyz}(xyz)^2 = 2\log_{xyz}(xyz) = 2.\end{aligned}$$

Answer: 2

CONCEPT

If the same number sits inside several logs in the denominators, flip every term. The common number moves into the base, and logs with a common base can be added.

(a) Exercise Q.64 I.I.F.T. 2005

Find

$$\frac{1}{\log_a bc + 1} + \frac{1}{\log_b ca + 1} + \frac{1}{\log_c ab + 1}.$$

Step 1. Absorb the "+1". Write $1 = \log_a a$. Then

$$\log_a bc + 1 = \log_a bc + \log_a a = \log_a(abc).$$

In the same way the other denominators are $\log_b(abc)$ and $\log_c(abc)$.

Step 2. Flip and add.

$$\log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc}(abc) = 1.$$

Answer: 1

(b) Exercise Q.57

Find $\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60}$.

$$\log_{60} 3 + \log_{60} 4 + \log_{60} 5 = \log_{60}(3 \times 4 \times 5) = \log_{60} 60 = 1.$$

Answer: 1

(c) Exercise Q.65

Find

$$\frac{1}{\log_{\left(\frac{p}{q}\right)} x} + \frac{1}{\log_{\left(\frac{q}{r}\right)} x} + \frac{1}{\log_{\left(\frac{r}{p}\right)} x}.$$

$$\begin{aligned}&\log_x \frac{p}{q} + \log_x \frac{q}{r} + \log_x \frac{r}{p} \\ &= \log_x \left(\frac{p}{q} \times \frac{q}{r} \times \frac{r}{p}\right) = \log_x 1 = 0.\end{aligned}$$

Answer: 0

(d) Extra

Show that

$$\frac{1}{\log_2 N} + \frac{1}{\log_3 N} + \cdots + \frac{1}{\log_{43} N} = \frac{1}{\log_{43!} N},$$

where $43! = 1 \times 2 \times 3 \times \cdots \times 43$.

Flip every term on the left:

$$\log_N 2 + \log_N 3 + \cdots + \log_N 43 = \log_N(2 \times 3 \times \cdots \times 43).$$

Multiplying by 1 changes nothing, so $2 \times 3 \times \cdots \times 43 = 43!$. Hence the left side is $\log_N(43!) = \frac{1}{\log_{43!} N}$.

(e) Extra

If $x = \log_a(bc)$, $y = \log_b(ca)$ and $z = \log_c(ab)$, prove that $xyz = x + y + z + 2$.

Step 1. Reuse part (a). It showed that

$$\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1} = 1.$$

Step 2. Clear the fractions by multiplying both sides by $(x+1)(y+1)(z+1)$:

$$(y+1)(z+1)+(x+1)(z+1)+(x+1)(y+1)$$
$$=(x+1)(y+1)(z+1).$$

Step 3. Expand both sides.

$$\text{Left} = xy+yz+zx+2(x+y+z)+3,$$
$$\text{Right} = xyz+xy+yz+zx+(x+y+z)+1.$$

Step 4. Cancel $xy+yz+zx$ from both sides:

$$2(x+y+z)+3 = xyz+(x+y+z)+1,$$

which gives $xyz = x+y+z+2$. ∎

Part C · Relations Between Logs

Questions 9–12: answers in letters instead of numbers

Q.9 ONE LOG IN TERMS OF ANOTHER

Exercise Q.53

If $\log_{12} 27 = a$, find $\log_6 16$ in terms of a.

SOLUTION

The idea. Both logs are built from the primes 2 and 3 only: $12 = 2^2\cdot 3$, $27 = 3^3$, $6 = 2\cdot 3$ and $16 = 2^4$. So everything can be written through one ratio,

$$t = \log_2 3 = \frac{\log 3}{\log 2}.$$

First find t from a, then use t to build $\log_6 16$.

Step 1. Write a using t.

$$a = \frac{\log 27}{\log 12} = \frac{3\log 3}{2\log 2+\log 3}.$$

Divide the top and the bottom by $\log 2$:

$$a = \frac{3t}{2+t}.$$

Step 2. Solve for t.

$$a(2+t) = 3t \;\Rightarrow\; 2a = 3t-at = t(3-a) \;\Rightarrow\; t = \frac{2a}{3-a}.$$

Step 3. Write the required log using t.

$$\log_6 16 = \frac{4\log 2}{\log 2+\log 3} = \frac{4}{1+t}.$$

Step 4. Substitute t.

$$\frac{4}{1+\dfrac{2a}{3-a}} = \frac{4}{\dfrac{3-a+2a}{3-a}} = \frac{4(3-a)}{3+a}.$$

Numerical check. With $a = \log_{12} 27 \approx 1.3263$,

$$\frac{4(3-a)}{3+a} \approx 1.5474 \quad\text{and}\quad \log_6 16 = \frac{1.2041}{0.7782} \approx 1.5474. \checkmark$$

Answer: $\log_6 16 = \dfrac{4(3-a)}{3+a}$

CONCEPT: THE ONE-RATIO METHOD

If every number in the problem is made of the same two primes, then every log in it depends only on the ratio of their logs. Find that ratio from the given value, then substitute it.

(a) Exercise Q.42 M.B.A. 2010

If $\log_{10} 2 = a$ and $\log_{10} 3 = b$, find $\log_5 12$.

The key fact is that $5 = \dfrac{10}{2}$, so $\log 5 = \log 10 - \log 2 = 1-a$. Also $\log 12 = \log(2^2\cdot 3) = 2a+b$. Therefore

$$\log_5 12 = \frac{\log 12}{\log 5} = \frac{2a+b}{1-a}.$$

Answer: $\dfrac{2a+b}{1-a}$

(b) Exercise Q.43

If $\log 2 = x$, $\log 3 = y$ and $\log 7 = z$, find $\log\left(4\sqrt[3]{63}\right)$.

Factorise: $63 = 3^2\times 7$, and a cube root is the power $\dfrac{1}{3}$.

$$\log\left(4\sqrt[3]{63}\right) = \log 4 + \frac{1}{3}\log 63$$
$$= 2\log 2 + \frac{1}{3}(2\log 3+\log 7)$$
$$= 2x+\frac{2}{3}y+\frac{1}{3}z.$$

Answer: $2x+\dfrac{2}{3}y+\dfrac{1}{3}z$

(c) Exercise Q.72 M.B.A. 2010

If $a = \log_8 225$ and $b = \log_2 15$, find a in terms of b.

Write $8 = 2^3$ and $225 = 15^2$, then use $\log_{a^q} x^p = \dfrac{p}{q}\log_a x$:

$$a = \log_{2^3} 15^2 = \frac{2}{3}\log_2 15 = \frac{2b}{3}.$$

Answer: $a = \dfrac{2b}{3}$

(d) Extra

If $\log_2 3 = a$ and $\log_3 5 = b$, find $\log_{10} 45$ in terms of a and b.

Work in base 2. By the chain rule, $\log_2 5 = \log_2 3\times\log_3 5 = ab$. Then

$$\log_2 45 = \log_2(9\times 5) = 2a+ab,$$
$$\log_2 10 = \log_2 2+\log_2 5 = 1+ab.$$
$$\therefore\ \log_{10} 45 = \frac{\log_2 45}{\log_2 10} = \frac{2a+ab}{1+ab}.$$

Answer: $\dfrac{a(2+b)}{1+ab}$

Q.10 RELATIONS BETWEEN UNKNOWN LOGS

Exercise Q.40 M.A.T. 2002

If $\log_a(ab) = x$, find $\log_b(ab)$.

SOLUTION

Step 1. Peel off the base. The number ab contains the base a, and $\log_a a = 1$:

$$\log_a(ab) = \log_a a+\log_a b = 1+\log_a b = x.$$

So $\log_a b = x-1$.

Step 2. Flip it. $\log_b a = \dfrac{1}{\log_a b} = \dfrac{1}{x-1}$.

Step 3. Peel off the base b in the same way.

$$\log_b(ab) = \log_b a+1 = \frac{1}{x-1}+1 = \frac{x}{x-1}.$$

Check. Take $a = 2$ and $b = 8$. Then $x = \log_2 16 = 4$, and

$$\log_8 16 = \frac{4}{3} = \frac{4}{4-1}. \checkmark$$

Answer: $\dfrac{x}{x-1}$

CONCEPT
Whenever the number inside a log contains the base as a factor, split it off: $\log_a(a \cdot N) = 1 + \log_a N$. This is much shorter than changing everything to base 10.

(a) Exercise Q.41

If $\log_a m = x$, find $\log_{\frac{1}{a}}\left(\dfrac{1}{m}\right)$.

$\dfrac{1}{a} = a^{-1}$ and $\dfrac{1}{m} = m^{-1}$, so

$$\log_{a^{-1}} m^{-1} = \frac{-1}{-1}\log_a m = x.$$

Answer: x

(b) Exercise Q.20

If $\log_x y = 100$ and $\log_2 x = 10$, find y.

The second fact gives $x = 2^{10}$. The first fact then gives

$$y = x^{100} = \left(2^{10}\right)^{100} = 2^{1000}.$$

Answer: $y = 2^{1000}$

(c) Exercise Q.13

If $\log_8 p = 25$ and $\log_2 q = 5$, how are p and q related?

$p = 8^{25} = \left(2^3\right)^{25} = 2^{75}$ and $q = 2^5$. Since $75 = 5 \times 15$,

$$p = 2^{75} = \left(2^5\right)^{15} = q^{15}.$$

Answer: $p = q^{15}$

(d) Exercise Q.66 C.D.S. 2003

If $\log_{10} 7 = a$, find $\log_{10}\left(\dfrac{1}{70}\right)$.

$$\log \frac{1}{70} = -\log 70 = -(\log 7 + \log 10) = -(a+1).$$

Answer: $-(1+a)$

(e) Extra

If $\log_a(ab) = x$, find $\log_{ab} b$ in terms of x.

Since $b = \dfrac{ab}{a}$,

$$\log_{ab} b = \log_{ab}(ab) - \log_{ab} a = 1 - \frac{1}{\log_a(ab)} = 1 - \frac{1}{x}.$$

Answer: $\dfrac{x-1}{x}$

Q.11 EQUAL POWERS

Exercise Q.69 M.B.A. 2008
If $a = b^2 = c^3 = d^4$, find $\log_a(abcd)$.

SOLUTION

Step 1. Write every letter as a power of a.

$$b^2 = a \Rightarrow b = a^{\frac{1}{2}},$$
$$c^3 = a \Rightarrow c = a^{\frac{1}{3}},$$
$$d^4 = a \Rightarrow d = a^{\frac{1}{4}}.$$

Step 2. Multiply (powers of the same base add):

$$abcd = a^1 \cdot a^{\frac{1}{2}} \cdot a^{\frac{1}{3}} \cdot a^{\frac{1}{4}} = a^{1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}}.$$

Step 3. Read off the log.

$$\log_a(abcd) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12+6+4+3}{12} = \frac{25}{12}.$$

Answer: $1 + \dfrac{1}{2} + \dfrac{1}{3} + \dfrac{1}{4} = \dfrac{25}{12}$

CONCEPT
When several expressions are equal, write every letter as a power of **one** of them, or of their common value k. Products then turn into sums of exponents.

(a) Exercise Q.67

If $a = b^x$, $b = c^y$ and $c = a^z$, find xyz.

Substitute one into the next until you return to a:

$$a = b^x = (c^y)^x = c^{xy} = (a^z)^{xy} = a^{xyz}.$$

So $a^1 = a^{xyz}$. For a positive $a \neq 1$, equal powers of a must have equal exponents, so $xyz = 1$.

With logs. $x = \log_b a$, $y = \log_c b$ and $z = \log_a c$ form a closed loop, so their product is 1 (Figure 3).

Answer: $xyz = 1$

(b) Exercise Q.32

If $a^x = b^y$, find $\dfrac{\log a}{\log b}$.

Take the log of both sides: $x \log a = y \log b$. Divide by $x \log b$:

$$\frac{\log a}{\log b} = \frac{y}{x}.$$

Answer: $\dfrac{y}{x}$

(c) Extra

If $2^x = 3^y = 6^{-z}$ (all non-zero), prove that $\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = 0$.

Call the common value k. Then

$$2 = k^{\frac{1}{x}}, \qquad 3 = k^{\frac{1}{y}}, \qquad 6 = k^{-\frac{1}{z}}.$$

Now use $2 \times 3 = 6$:

$$k^{\frac{1}{x}} \cdot k^{\frac{1}{y}} = k^{-\frac{1}{z}} \Rightarrow \frac{1}{x} + \frac{1}{y} = -\frac{1}{z}.$$

Moving $-\dfrac{1}{z}$ to the left gives $\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = 0$. ∎

(d) Extra

If $a^x = b^y = c^z$ and $b^2 = ac$, prove that $y = \dfrac{2xz}{x+z}$.

With common value k: $a = k^{\frac{1}{x}}$, $b = k^{\frac{1}{y}}$ and $c = k^{\frac{1}{z}}$. Then $b^2 = ac$ becomes

$$k^{\frac{2}{y}} = k^{\frac{1}{x}+\frac{1}{z}} \Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z} = \frac{x+z}{xz}.$$

Turning both sides upside down and multiplying by 2 gives $y = \dfrac{2xz}{x+z}$. (So x, y, z are in harmonic progression.) ∎

Q.12 CONDITIONS AND IDENTITIES

Exercise Q.60

If $\log x + \log y = \log(x+y)$, express y in terms of x.

SOLUTION

Step 1. Combine the left side. $\log x + \log y = \log(xy)$, so

$$\log(xy) = \log(x+y).$$

Step 2. Drop the logs. Each positive number has exactly one log (Figure 1), so equal logs mean equal numbers:

$$xy = x + y.$$

Step 3. Make y the subject. Collect the y terms on one side:

$$xy - y = x \;\Rightarrow\; y(x-1) = x \;\Rightarrow\; y = \frac{x}{x-1}.$$

Since y must be positive, this works only for $x > 1$.

Check. Put $x = 2$; then $y = 2$, and indeed

$$\log 2 + \log 2 = \log 4 = \log(2+2). \checkmark$$

Answer: $y = \dfrac{x}{x-1}$

(a) Exercise Q.61 M.B.A. 2007

If $\log \dfrac{a}{b} + \log \dfrac{b}{a} = \log(a+b)$, find the condition on a and b.

The left side is $\log\left(\dfrac{a}{b} \times \dfrac{b}{a}\right) = \log 1 = 0$. So $\log(a+b) = 0$, which means $a + b = 10^0 = 1$.

Answer: $a + b = 1$

(b) Exercise Q.56 R.R.B. 2005

When is $\dfrac{1}{2}(\log x + \log y) = \log \dfrac{x+y}{2}$?

The left side is $\dfrac{1}{2}\log(xy) = \log\sqrt{xy}$. Dropping the logs,

$$\sqrt{xy} = \frac{x+y}{2} \;\Rightarrow\; x + y - 2\sqrt{x}\sqrt{y} = 0 \;\Rightarrow\; \left(\sqrt{x} - \sqrt{y}\right)^2 = 0.$$

A square is zero only when the number inside is zero, so $\sqrt{x} = \sqrt{y}$, that is, $x = y$.

Answer: $x = y$

CONCEPT

$\sqrt{xy}$ is the geometric mean (GM) and $\dfrac{x+y}{2}$ is the arithmetic mean (AM). Always AM $\geq$ GM, with equality only when the two numbers are equal.

(c) Extra

If $\log \dfrac{a+b}{3} = \dfrac{1}{2}(\log a + \log b)$, show that $\dfrac{a}{b} + \dfrac{b}{a} = 7$.

As in (b), the right side is $\log\sqrt{ab}$. Dropping logs and squaring:

$$\frac{a+b}{3} = \sqrt{ab} \;\Rightarrow\; (a+b)^2 = 9ab.$$

Expanding, $a^2 + 2ab + b^2 = 9ab$, so $a^2 + b^2 = 7ab$. Dividing by ab gives $\dfrac{a}{b} + \dfrac{b}{a} = 7$. ∎

(d) Extra

If $x^2 + y^2 = 6xy$, prove that $2\log(x+y) = \log x + \log y + 3\log 2$.

Build the square on the left:

$$(x+y)^2 = x^2 + y^2 + 2xy = 6xy + 2xy = 8xy.$$

Take logs: $2\log(x+y) = \log 8 + \log x + \log y = 3\log 2 + \log x + \log y$. ∎

CONCEPT

Questions (c) and (d) are the same idea seen from opposite ends. An identity like $x^2 + y^2 = k\,xy$ always becomes $(x+y)^2 = (k+2)\,xy$, which turns into a log statement.

Part D · Logarithmic Equations

Questions 13–16: finding the unknown, and checking it

Q.13 EQUATIONS WITH MIXED BASES

Exercise Q.70 **M.B.A. 2010**

Solve

$$\log_3 x + \log_9 x^2 + \log_{27} x^3 = 9.$$

SOLUTION

Step 1. Bring every term to base 3. The bases 9 and 27 are powers of 3 ($9 = 3^2$, $27 = 3^3$). Use $\log_{a^q} x^p = \dfrac{p}{q}\log_a x$:

$$\log_{3^2} x^2 = \frac{2}{2}\log_3 x = \log_3 x, \qquad \log_{3^3} x^3 = \frac{3}{3}\log_3 x = \log_3 x.$$

Step 2. The equation becomes very simple.

$$\log_3 x + \log_3 x + \log_3 x = 9 \;\Rightarrow\; 3\log_3 x = 9 \;\Rightarrow\; \log_3 x = 3.$$

Step 3. $x = 3^3 = 27$, which is positive, so it is allowed.

Answer: $x = 27$

CONCEPT

If the bases are different powers of one number, rewrite every log in that number's base. The equation then has a single unknown, $\log_3 x$, and becomes ordinary algebra.

(a) Exercise Q.44

Solve $\log_4 x + \log_2 x = 6$.

$\log_4 x = \log_{2^2} x = \dfrac{1}{2}\log_2 x$. Writing $t = \log_2 x$:

$$\frac{1}{2}t + t = 6 \;\Rightarrow\; \frac{3}{2}t = 6 \;\Rightarrow\; t = 4 \;\Rightarrow\; x = 2^4 = 16.$$

Answer: $x = 16$

(b) Exercise Q.48

Solve $\log_8 x + \log_8 \dfrac{1}{6} = \dfrac{1}{3}$.

Combine: $\log_8 \dfrac{x}{6} = \dfrac{1}{3}$. Power form: $\dfrac{x}{6} = 8^{\frac{1}{3}} = \sqrt[3]{8} = 2$, so $x = 12$.

Answer: $x = 12$

(c) Extra

Solve $\log_2 x + \log_4 x + \log_{16} x = \dfrac{21}{4}$.

In base 2, with $t = \log_2 x$: $\log_4 x = \dfrac{t}{2}$ and $\log_{16} x = \dfrac{t}{4}$. So

$$t + \frac{t}{2} + \frac{t}{4} = \frac{7t}{4} = \frac{21}{4} \;\Rightarrow\; t = 3 \;\Rightarrow\; x = 2^3 = 8.$$

Answer: $x = 8$

(d) Extra

Solve

$$\log_x 2 \cdot \log_{\frac{x}{16}} 2 = \log_{\frac{x}{64}} 2.$$

Step 1. Move x out of the bases. Flip each log and work in base 2, with $t = \log_2 x$:

$$\log_x 2 = \frac{1}{\log_2 x} = \frac{1}{t}, \qquad \log_{\frac{x}{16}} 2 = \frac{1}{\log_2 \frac{x}{16}} = \frac{1}{t-4},$$

$$\log_{\frac{x}{64}} 2 = \frac{1}{t-6}.$$

Step 2. Solve for t.

$$\frac{1}{t(t-4)} = \frac{1}{t-6} \;\Rightarrow\; t - 6 = t^2 - 4t \;\Rightarrow\; t^2 - 5t + 6 = 0.$$

So $(t-2)(t-3) = 0$, giving $t = 2$ or $t = 3$.

Step 3. Back to x and check the bases. $x = 2^2 = 4$ gives bases 4, $\frac{1}{4}$, $\frac{1}{16}$; $x = 2^3 = 8$ gives 8, $\frac{1}{2}$, $\frac{1}{8}$. All are positive and none is 1, so both roots are valid.

Answer: $x = 4$ **or** $x = 8$

Q.14 DOMAIN FIRST, THEN FALSE ROOTS

Exercise Q.55 · M.B.A. 2007

Solve

$$\log_5 (x^2 + x) - \log_5 (x+1) = 2.$$

SOLUTION

Step 1. Write the domain. Each number inside a log must be positive.

- $x + 1 > 0$, so $x > -1$.
- $x^2 + x = x(x+1) > 0$. A product is positive when both factors have the same sign, so $x < -1$ or $x > 0$.

Both conditions hold together only for $x > 0$ (Figure 4).

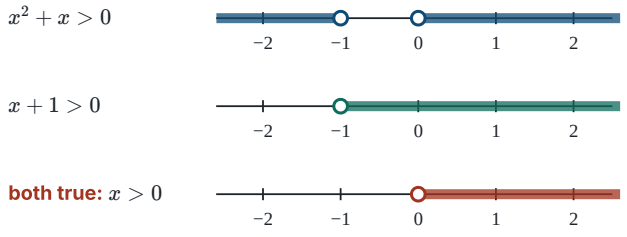


Figure 4. Finding the domain: keep only the part of the line where every condition holds.

Step 2. Combine the logs. A difference of logs is the log of a quotient:

$$\log_5 \frac{x(x+1)}{x+1} = 2 \;\Rightarrow\; \log_5 x = 2.$$

(Cancelling $x + 1$ is allowed because $x + 1 \neq 0$ in the domain.)

Step 3. Drop the log. $x = 5^2 = 25$.

Step 4. Check. $25 > 0$, so it lies in the domain.

Answer: $x = 25$

CONCEPT: THE FOUR-STEP ROUTINE
(1) Write the domain. (2) Combine the logs into one log on each side. (3) Drop the log, using the power form or "equal logs mean equal numbers". (4) Solve, and test every root against the domain.

(a) Solved Ex. 8 · M.B.A. 2002

Solve

$$\log_{10} 3 + \log_{10}(4x+1) = \log_{10}(x+1) + 1.$$

Domain: $4x + 1 > 0$ and $x + 1 > 0$, so $x > -\dfrac{1}{4}$.

Make every term a log: $1 = \log_{10} 10$. Then

$$\log\left[3(4x+1)\right] = \log\left[10(x+1)\right].$$

Drop the logs: $12x + 3 = 10x + 10$, so $2x = 7$ and $x = \dfrac{7}{2}$, which is in the domain.

Answer: $x = \dfrac{7}{2}$

TRAP
The "+1" on the right is $\log 10$, not $\log 1$ (which is 0).

(b) Solved Ex. 7 · R.R.B. 2006

Solve $\log_{10}\left(x^2 - 6x + 45\right) = 2$.

Power form: $x^2 - 6x + 45 = 10^2 = 100$, so $x^2 - 6x - 55 = 0$. Two numbers with product -55 and sum -6 are -11 and 5:

$$(x - 11)(x + 5) = 0 \;\Rightarrow\; x = 11 \text{ or } x = -5.$$

For both values the number inside the log is 100, which is positive, so **both** roots are valid.

Answer: $x = 11$ **or** $x = -5$

NOTE
A negative value of x is fine here. Only the number **inside** the log has to be positive, and x never appears alone inside a log.

(c) Exercise Q.54

Solve $\log_{10} 5 + \log_{10}(5x+1) = \log_{10}(x+5) + 1$.

As in (a), write $1 = \log 10$ and drop the logs:

$$5(5x+1) = 10(x+5)$$
$$\Rightarrow\; 25x + 5 = 10x + 50 \;\Rightarrow\; 15x = 45.$$

So $x = 3$; the numbers inside the logs are 16 and 8, both positive.

Answer: $x = 3$

(d) Exercise Q.45

Solve $\log_{10}\left(x^2 - 6x + 10\right) = 0$.

Power form: $x^2 - 6x + 10 = 10^0 = 1$, so $x^2 - 6x + 9 = 0$, that is, $(x-3)^2 = 0$.

Answer: $x = 3$

(e) Extra

Solve $\log x + \log(x-3) = 1$.

Domain: $x > 0$ and $x - 3 > 0$, so $x > 3$.

Solve: $\log[x(x-3)] = \log 10$, so

$$x^2 - 3x - 10 = 0 \Rightarrow (x-5)(x+2) = 0$$
$$\Rightarrow x = 5 \text{ or } x = -2.$$

Check: $x = -2$ is outside the domain, because $\log(-2)$ does not exist. Reject it.

Answer: $x = 5$ **only**

TRAP
The algebra produced a false root, $x = -2$. Only the domain check catches it, so never skip Step 4.

(f) Extra

Solve and compare: (i) $\log x^2 = 2$; (ii) $2\log x = 2$.

(i) $x^2 = 10^2 = 100$, so $x = 10$ or $x = -10$. Both are valid, since $x^2 = 100 > 0$ in each case.

(ii) $\log x = 1$ needs $x > 0$, so $x = 10$ only.

Answer: (i) $x = \pm 10$ **(ii)** $x = 10$

CONCEPT
$\log x^2 = 2\log x$ is true only when $x > 0$. In general $\log x^2 = 2\log|x|$. Using the power law without thinking loses the root -10 in (i).

Q.15 RADICALS, SYSTEMS, 2^x INSIDE A LOG

Exercise Q.71 M.B.A. 2009

Solve

$$\log_7 \log_5 \left(\sqrt{x+5} + \sqrt{x}\right) = 0.$$

SOLUTION

Step 1. Outer log. $\log_7 A = 0$ means $A = 7^0 = 1$. So

$$\log_5 \left(\sqrt{x+5} + \sqrt{x}\right) = 1.$$

Step 2. Next log. $\sqrt{x+5} + \sqrt{x} = 5^1 = 5$.

Step 3. Solve the root equation. Keep one root alone on one side, then square:

$$\sqrt{x+5} = 5 - \sqrt{x} \Rightarrow x + 5 = 25 - 10\sqrt{x} + x.$$

The x cancels on both sides, leaving $10\sqrt{x} = 20$. So $\sqrt{x} = 2$ and $x = 4$.

Step 4. Check. $\sqrt{9} + \sqrt{4} = 3 + 2 = 5$, then $\log_5 5 = 1$ and $\log_7 1 = 0$. ✓

Answer: $x = 4$

CONCEPT
Before squaring a root equation, isolate one root. Otherwise the cross term $2\sqrt{x+5}\sqrt{x}$ appears and the work doubles. Squaring can also create false roots, so always check.

(a) Exercise Q.46

Solve the pair of equations

$$\log_{10} x + \log_{10} y = 3, \qquad \log_{10} x - \log_{10} y = 1.$$

Treat $X = \log x$ and $Y = \log y$ as the unknowns:

$$X + Y = 3, \quad X - Y = 1 \Rightarrow 2X = 4.$$

So $X = 2$ and $Y = 1$, which gives $x = 10^2 = 100$ and $y = 10^1 = 10$.

Answer: $x = 100, y = 10$

(b) Extra

Solve the pair of equations

$$\log_x y + \log_y x = \frac{5}{2}, \qquad xy = 27.$$

Step 1. Let $t = \log_x y$; then $\log_y x = \frac{1}{t}$. Multiply $t + \frac{1}{t} = \frac{5}{2}$ by $2t$:

$$2t^2 - 5t + 2 = 0 \Rightarrow (2t-1)(t-2) = 0 \Rightarrow t = 2 \text{ or } t = \frac{1}{2}.$$

Step 2. If $t = 2$, then $y = x^2$, and $xy = x^3 = 27$ gives $x = 3$, $y = 9$.

If $t = \frac{1}{2}$, then $y = x^{\frac{1}{2}}$, and $xy = x^{\frac{3}{2}} = 27$. So

$$x = 27^{\frac{2}{3}} = \left(\sqrt[3]{27}\right)^2 = 9, \qquad y = 3.$$

Answer: $(x, y) = (3, 9)$ **or** $(9, 3)$

(c) Extra

Solve $\log_2(9 - 2^x) = 3 - x$.

Step 1. Power form.

$$9 - 2^x = 2^{3-x} = \frac{8}{2^x}.$$

Step 2. Substitute $t = 2^x$ (always positive) and multiply by t:

$$9t - t^2 = 8 \Rightarrow t^2 - 9t + 8 = 0 \Rightarrow (t-1)(t-8) = 0.$$

Step 3. $2^x = 1$ gives $x = 0$; $2^x = 8$ gives $x = 3$.

Check:

$$x = 0: \ \log_2 8 = 3 = 3 - 0 \ ✓$$
$$x = 3: \ \log_2 1 = 0 = 3 - 3 \ ✓$$

Answer: $x = 0$ **or** $x = 3$

Q.16 QUADRATICS IN DISGUISE; AP OF LOGS

Exercise Q.91 IIFT 2012

If $\log 3$, $\log(3^x - 2)$ and $\log(3^x + 4)$ are in arithmetic progression, find x.

SOLUTION

Step 1. Write the AP condition. Three numbers p, q, r are in AP when $q - p = r - q$, that is, $2q = p + r$:

$$2\log(3^x-2)=\log 3+\log(3^x+4).$$

Step 2. Drop the logs. The left side is $\log(3^x-2)^2$ and the right side is $\log[3(3^x+4)]$, so

$$(3^x-2)^2=3(3^x+4).$$

Step 3. Substitute $t=3^x$. This needs $t>2$, so that $\log(3^x-2)$ exists.

$$t^2-4t+4=3t+12 \Rightarrow t^2-7t-8=0$$
$$\Rightarrow (t-8)(t+1)=0.$$

$t=-1$ is impossible, because 3^x is always positive. So $3^x=8$.

Step 4. $x=\log_3 8=3\log_3 2\approx 1.893$.

Check. $3^x-2=6$ and $3^x+4=12$. The logs $\log 3$, $\log 6$, $\log 12$ go up by $\log 2$ each time. ✓

Answer: $x=\log_3 8$

NOTE ON THE PDF
The PDF's option (c), printed as "$\log 2^3$", stands for $\log_3 2^3=\log_3 8$. The PDF's printed working for this question is muddled; the steps above are the correct ones.

CONCEPT
Logs in AP $\Longleftrightarrow$ numbers in GP. Here 3, 6, 12 is a GP with ratio 2, so their logs form an AP. This gives a quick way to check the answer.

(a) Extra

If $\log 2$, $\log(2^x-1)$ and $\log(2^x+3)$ are in AP, find x.

With $t=2^x$, the AP condition gives $(t-1)^2=2(t+3)$:

$$t^2-2t+1=2t+6 \Rightarrow t^2-4t-5=0$$
$$\Rightarrow (t-5)(t+1)=0.$$

So $2^x=5$, and the numbers are 2, 4, 8, a GP. ✓

Answer: $x=\log_2 5$

(b) Extra

Solve $x^{\log x}=100x$ (logs to base 10).

Step 1. Take the log of both sides. The power law brings the exponent down:

$$\log x\cdot\log x=\log 100+\log x.$$

Step 2. With $t=\log x$: $t^2-t-2=0$, so $(t-2)(t+1)=0$ and $t=2$ or $t=-1$.

Step 3. $x=10^2=100$ or $x=10^{-1}=\dfrac{1}{10}$.

Check $x=\dfrac{1}{10}$: $\left(\dfrac{1}{10}\right)^{-1}=10$ and $100\times\dfrac{1}{10}=10$. ✓

Answer: $x=100$ **or** $x=\dfrac{1}{10}$

(c) Extra

Solve

$$\log_2(4^x+4)=x+\log_2(2^{x+1}-3).$$

Step 1. Make x a log. $x=\log_2 2^x$, so the right side is $\log_2[2^x(2^{x+1}-3)]$.

Step 2. Drop the logs and put $t=2^x$ (so $4^x=t^2$ and $2^{x+1}=2t$):

$$t^2+4=t(2t-3) \Rightarrow t^2-3t-4=0 \Rightarrow (t-4)(t+1)=0.$$

Step 3. $2^x=4$, so $x=2$. Check: $2^3-3=5>0$, and both sides equal $\log_2 20$. ✓

Answer: $x=2$

Part E · Working with Log Values

Questions 17–19: given logs, the bar form, and counting digits

Q.17 VALUES FROM $\log 2$, $\log 3$, $\log 7$

Exercise Q.80 M.A.T. 2002

If $\log 2=0.3010$ and $\log 3=0.4771$, find $\log_5 512$.

SOLUTION

Step 1. Change to base 10. The given values are base-10 logs, so

$$\log_5 512=\frac{\log 512}{\log 5}.$$

Step 2. Top. $512=2^9$, so $\log 512=9\log 2=9\times 0.3010=2.7090$.

Step 3. Bottom. $5=\dfrac{10}{2}$, so $\log 5=1-\log 2=1-0.3010=0.6990$.

Step 4. Divide.

$$\log_5 512=\frac{2.7090}{0.6990}=\frac{2709}{699}\approx 3.876.$$

Answer: ≈ 3.876

CONCEPT
With only $\log 2$ and $\log 3$ you can find the log of any number built from 2, 3 and 5, because $\log 5=1-\log 2$. Here $\log 3$ was not needed at all; examiners often give extra data to distract you.

(a) Solved Ex. 11 & 12

Using $\log 2=0.30103$ in (i) and $\log 2=0.3010$, $\log 3=0.4771$ in (ii) and (iii), find (i) $\log 50$; (ii) $\log 25$; (iii) $\log 4.5$.

(i) $\log 50=\log\dfrac{100}{2}=2-0.30103=1.69897$.

(ii) $\log 25=\log\dfrac{100}{4}=2-2(0.3010)=1.3980$.

(iii) $\log 4.5=\log\dfrac{9}{2}=2(0.4771)-0.3010=0.6532$.

Answer: 1.69897, 1.3980, 0.6532

(b) Exercise Q.81

If $\log 3=0.4771$ and $\log 7=0.8451$, find $\log 23\dfrac{1}{3}$.

$23\dfrac{1}{3}=\dfrac{70}{3}$ and $\log 70=\log 7+\log 10=1.8451$. So

$$\log\frac{70}{3}=1.8451-0.4771=1.3680.$$

Answer: 1.368

(c) Exercise Q.83

If $\log 2=0.3010$ and $\log 7=0.8451$, find $\log 2.8$.

$2.8 = \dfrac{28}{10} = \dfrac{2^2 \times 7}{10}$, so

$$\log 2.8 = 2(0.3010) + 0.8451 - 1 = 0.4471.$$

Answer: 0.4471

(d) Exercise Q.77 & Q.73

(i) If $\log 3 = 0.477$ and $(1000)^x = 3$, find x. (ii) If $\log 27 = 1.431$, find $\log 9$.

(i) Take logs: $x \log 1000 = \log 3$, that is, $3x = 0.477$, so $x = 0.159$.

(ii) $\log 27 = 3 \log 3 = 1.431$ gives $\log 3 = 0.477$, so $\log 9 = 2 \log 3 = 0.954$.

Answer: (i) 0.159 (ii) 0.954

(e) Exercise Q.74 & Q.79

(i) If $\log_{10} 2 = 0.3010$, find $\log_2 10$. (ii) If $\log 20 = 1.3010$ and $\log 30 = 1.4771$, find $\log 60000$.

(i) Swap base and number: $\log_2 10 = \dfrac{1}{\log_{10} 2} = \dfrac{1}{0.3010} = \dfrac{1000}{301} \approx 3.322$.

(ii) $60000 = 20 \times 30 \times 100$, so $\log 60000 = 1.3010 + 1.4771 + 2 = 4.7781$.

Answer: (i) $\dfrac{1000}{301}$ (ii) 4.7781

(f) Extra

Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, solve $5^x = 12$.

Take logs of both sides: $x \log 5 = \log 12$. Here $\log 12 = 2(0.3010) + 0.4771 = 1.0791$ and $\log 5 = 0.6990$, so

$$x = \frac{1.0791}{0.6990} \approx 1.544.$$

Answer: $x \approx 1.544$

Q.18 CHARACTERISTIC, MANTISSA, BAR FORM

Exercise Q.84

If $\log(0.57) = \overline{1}.756$, find

$$\log 57 + \log(0.57)^3 + \log \sqrt{0.57}.$$

SOLUTION

Step 1. Read the bar form. $\overline{1}.756$ means $-1 + 0.756 = -0.244$. The bar sits only on the whole-number part (the **characteristic**), which is negative. The decimal part (the **mantissa**) stays positive.

Step 2. $\log 57$. $57 = 0.57 \times 100$, so

$$\log 57 = \log 0.57 + 2 = -1 + 0.756 + 2 = 1.756.$$

The digits 5 and 7 are the same as in 0.57, so the mantissa .756 does not change; only the characteristic does.

Step 3. $\log(0.57)^3 = 3 \times (-0.244) = -0.732$.

Step 4. $\log \sqrt{0.57} = \dfrac{1}{2} \times (-0.244) = -0.122$.

Step 5. Add. $1.756 - 0.732 - 0.122 = 0.902$.

Answer: 0.902

The same work in bar form (as done with log tables):

- $3 \times \overline{1}.756 = \overline{3} + 2.268$. Carry the 2 to the characteristic: $-3 + 2 = -1$, giving $\overline{1}.268$.
- To halve $\overline{1}.756$, first make the characteristic even: $\overline{1}.756 = \overline{2} + 1.756$. Half of that is $\overline{1} + 0.878 = \overline{1}.878$.
- Sum: $1.756 + \overline{1}.268 + \overline{1}.878 = -1 + 1.902 = 0.902$. ✓

CONCEPT

Characteristic: the whole-number part of a log, which may be negative. Mantissa: the decimal part, always between 0 and 1. Numbers with the same digits (57, 5.7, 0.57, 0.0057) have the same mantissa.

(a) Exercise Q.85 M.B.A. 2009

The logarithm of a number is -3.153. Find its characteristic and mantissa.

The mantissa must be positive, so go down to the next whole number **below** -3.153, which is -4, and add the positive remainder:

$$-3.153 = -4 + 0.847 = \overline{4}.847.$$

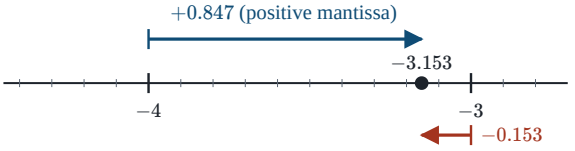


Figure 5. Writing -3.153 as -4 plus a positive part: step to -4, then forward by 0.847.

Answer: characteristic -4, mantissa 0.847

TRAP

"Characteristic -3, mantissa -0.153" is wrong: a mantissa is never negative.

(b) Important Facts, IV

Find the characteristic of the log of 348.25, 46.583, 9.2193, 0.6173, 0.03125 and 0.00125.

The characteristic is the power of 10 just below the number:

Number	lies between	characteristic
348.25	10^2 and 10^3	2
46.583	10^1 and 10^2	1
9.2193	10^0 and 10^1	0
0.6173	10^{-1} and 10^0	$\overline{1}$
0.03125	10^{-2} and 10^{-1}	$\overline{2}$
0.00125	10^{-3} and 10^{-2}	$\overline{3}$

This gives the PDF's two rules. For a number above 1, the characteristic is one less than the number of digits before the decimal point. For a number below 1, it is negative and one more than the number of zeros right after the decimal point.

(c) Extra

Given $\log 4.32 = 0.6355$, find $\log 43200$ and $\log 0.000432$.

$43200 = 4.32 \times 10^4$, so $\log 43200 = 4 + 0.6355 = 4.6355$.

$0.000432 = 4.32 \times 10^{-4}$, so $\log 0.000432 = -4 + 0.6355 = \overline{4}.6355$, which is -3.3645.

Answer: 4.6355 and $\overline{4}.6355$

(d) Extra

Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 0.0072$ in bar form.

$$0.0072 = \frac{72}{10000} = 2^3 \times 3^2 \times 10^{-4}, \text{ so}$$

$$\log 0.0072 = 3(0.3010) + 2(0.4771) - 4$$
$$= 1.8572 - 4 = \overline{3}.8572.$$

Check with the rule: 0.0072 has two zeros right after the point, so the characteristic is $-(2+1) = -3$. ✓

Answer: $\overline{3}.8572$

(e) Extra

If $\log x = \overline{2}.4771$ and $\log 3 = 0.4771$, find x.

$\log x = -2 + 0.4771 = \log 10^{-2} + \log 3 = \log\left(3 \times 10^{-2}\right)$, so $x = 0.03$.

Answer: $x = 0.03$

Q.19 NUMBER OF DIGITS; ZEROS AFTER THE POINT

Exercise Q.90 · A.A.O. Exam 2009

How many digits does $4^9 \times 5^{17}$ have?

SOLUTION 1: PAIR EACH 2 WITH A 5

$4^9 = \left(2^2\right)^9 = 2^{18}$. Each pair 2×5 makes a 10:

$$2^{18} \times 5^{17} = 2 \times \left(2^{17} \times 5^{17}\right) = 2 \times 10^{17}.$$

That is the digit 2 followed by 17 zeros, so it has $1 + 17 = 18$ digits.

SOLUTION 2: WITH LOGS

$$\log N = 18 \log 2 + 17 \log 5$$
$$= 18(0.30103) + 17(0.69897)$$
$$= 5.41854 + 11.88249 = 17.30103.$$

The characteristic is 17, so the number of digits is $17 + 1 = 18$.

Answer: 18 digits

N lies in	$\log N$ lies in	digits
$1 \le N < 10$	0 to 1	1
$10 \le N < 100$	1 to 2	2
$100 \le N < 1000$	2 to 3	3
$10^c \le N < 10^{c+1}$	c to $c+1$	$c+1$

Figure 6. The digit ladder: a characteristic of c means $c+1$ digits.

CONCEPT

If $\log N = c + m$, with c a whole number and $0 \le m < 1$, then $10^c \le N < 10^{c+1}$. A number from 10^c up to (not including) 10^{c+1} has exactly $c+1$ digits.

(a) Solved Ex. 13 · M.A.T. 2005

If $\log 2 = 0.30103$, find the number of digits in 2^{56}.

$\log 2^{56} = 56 \times 0.30103 = 16.85768$. The characteristic is 16, so there are 17 digits.

Answer: 17 digits

(b) Exercise Q.86 & Q.87 · Hotel Management 2009

With $\log 2 = 0.30103$, find the number of digits in (i) 2^{64}; (ii) 4^{50}.

(i) $64 \times 0.30103 = 19.26592$; the characteristic is 19, so 20 digits.

(ii) $4^{50} = 2^{100}$, and $100 \times 0.30103 = 30.103$; the characteristic is 30, so 31 digits.

Answer: (i) 20 (ii) 31

(c) Exercise Q.88

With $\log 2 = 0.30103$, find the number of digits in 5^{20}.

$\log 5 = 1 - 0.30103 = 0.69897$, and $20 \times 0.69897 = 13.9794$. The characteristic is 13, so 14 digits.

Answer: 14 digits

(d) Exercise Q.89

With $\log 2 = 0.30103$ and $\log 3 = 0.47712$, find the number of digits in 6^{20}.

$\log 6 = 0.30103 + 0.47712 = 0.77815$, and $20 \times 0.77815 = 15.563$. The characteristic is 15, so 16 digits.

Answer: 16 digits

(e) Extra

How many zeros are there between the decimal point and the first non-zero digit of $\left(\frac{1}{2}\right)^{50}$?

$\log\left(\frac{1}{2}\right)^{50} = -50 \times 0.30103 = -15.0515$. Make the mantissa positive:

$$-15.0515 = -16 + 0.9485 = \overline{16}.9485.$$

The number lies between 10^{-16} and 10^{-15}, so it looks like $0.000\ldots0d\ldots$ with 15 zeros after the point. In general, a characteristic of $-n$ means $n-1$ zeros right after the decimal point (compare 0.03125 in Q.18(b): characteristic $\overline{2}$, one zero).

Answer: 15 zeros

(f) Extra

How many digits does $8^{10} \times 5^{25}$ have?

$8^{10} = 2^{30}$. Pair 25 of the 2s with the 25 fives:

$$2^{30} \times 5^{25} = 2^5 \times \left(2^{25} \times 5^{25}\right) = 32 \times 10^{25}.$$

That is "32" followed by 25 zeros: $2 + 25 = 27$ digits.

Answer: 27 digits

Part F · Traps, Comparisons and Inequalities

Question 20: where most marks are lost

Q.20 TRAPS, COMPARISONS, INEQUALITIES

Exercise Q.24 M.B.A. 2003

Which of the following statements is **not** correct?

(a) $\log_{10} 10 = 1$
(b) $\log(2 + 3) = \log(2 \times 3)$
(c) $\log_{10} 1 = 0$
(d) $\log(1 + 2 + 3) = \log 1 + \log 2 + \log 3$

SOLUTION

Test each statement against the definition and the laws.

- **(a) True,** because $10^1 = 10$.
- **(b) False.** $\log(2 + 3) = \log 5 \approx 0.699$, but $\log(2 \times 3) = \log 6 \approx 0.778$. There is **no** law for the log of a sum.
- **(c) True,** because $10^0 = 1$.
- **(d) True, but only by luck.** $1 + 2 + 3 = 6 = 1 \times 2 \times 3$, so $\log(1 + 2 + 3) = \log(1 \times 2 \times 3) = \log 1 + \log 2 + \log 3$.

Answer: (b)

TRAP

(d) looks wrong because it seems to use "log of a sum = sum of logs". It is true only because 1, 2, 3 have the same sum and product. Always test with the actual numbers.

(a) Extra

True or false? Give a reason for each.

Statement	T/F	Reason
$\dfrac{\log_2 8}{\log_2 4} = \log_2 2$	F	Left side is $\dfrac{3}{2}$ (it is $\log_4 8$); right side is 1.
$\log a^2 = (\log a)^2$	F	$\log a^2 = 2 \log a$. For $a = 10$: 2 against 1.
$\log_3 5 \times \log_5 7 = \log_3 7$	T	The chain rule of Q.7.
$\log(2 + 8) = \log 2 + \log 8$	F	Left side is $\log 10 = 1$; right side is $\log 16 \approx 1.204$.

CONCEPT

A quotient of two logs is a **change of base**, $\dfrac{\log_c x}{\log_c y} = \log_y x$. It is not the log of a quotient, $\log \dfrac{x}{y} = \log x - \log y$.

(b) Extra

Which is greater, $\log_2 3$ or $\log_3 5$?

The idea. Find a simple number that lies between them. Try $\dfrac{3}{2}$. Both bases are above 1, so both logs are increasing (Figure 1), and we can compare powers instead of logs:

- $\log_2 3 > \dfrac{3}{2}$, because $2^{\frac{3}{2}} = \sqrt{8} \approx 2.83$ is less than 3;
- $\log_3 5 < \dfrac{3}{2}$, because $3^{\frac{3}{2}} = \sqrt{27} \approx 5.20$ is more than 5.

So $\log_3 5 < \dfrac{3}{2} < \log_2 3$. (In fact $\log_2 3 \approx 1.585$ and $\log_3 5 \approx 1.465$.)

Answer: $\log_2 3$ is greater

(c) Extra

Solve $\log_2(x - 3) < 2$.

Domain: $x - 3 > 0$, so $x > 3$.

Remove the log. The base 2 is above 1, so the curve rises (Figure 1) and the inequality keeps its direction:

$$x - 3 < 2^2 = 4 \;\Rightarrow\; x < 7.$$

Answer: $3 < x < 7$

(d) Extra

Solve

$$\log_{0.5}(2x - 1) > \log_{0.5}(x + 2).$$

Domain: $2x - 1 > 0$ gives $x > \dfrac{1}{2}$, and $x + 2 > 0$ gives $x > -2$. Together, $x > \dfrac{1}{2}$.

Remove the log. The base 0.5 is below 1, so the curve falls (the red curve in Figure 1): a **bigger** log comes from a **smaller** number. The inequality therefore turns round:

$$2x - 1 < x + 2 \;\Rightarrow\; x < 3.$$

Combine with the domain: $\dfrac{1}{2} < x < 3$.

Check. At $x = 1$: $\log_{0.5} 1 = 0$ and $\log_{0.5} 3 \approx -1.585$, so $0 > -1.585$ holds. ✓ At $x = 4$: $\log_{0.5} 7 \approx -2.81$ is not greater than $\log_{0.5} 6 \approx -2.58$, so 4 is correctly left out.

Answer: $\dfrac{1}{2} < x < 3$

TRAP

Forgetting to turn the inequality round gives $x > 3$, which is exactly the wrong set. When the base is between 0 and 1, the sign always flips.